$\sim \begin{pmatrix} 1 & -4 & 7 & 14 \\ 0 & 20 & -23 & -29 \\ 0 & 20 & -23 & -93 \end{pmatrix}$	$R_2 \to R_2 - 3R_1$ $R_3 \to R_3 - 7R_1$
$\sim \begin{pmatrix} 1 & -4 & 7 & 14 \\ 0 & 20 & -23 & -29 \\ 0 & 0 & 0 & 64 \end{pmatrix}$	$R_3 \to R_3 - R_2$
$\rho(A) = 2,\ \rho([A,B]) = 3$	

The last equivalent matrix is in the echelon form. $[A, B]$ has 3 non-zero rows and $[A]$ has 2 non-zero rows.

$\therefore \quad \rho([A, B]) = 3, \qquad \rho(A) = 2$

$\rho(A) \neq \rho([A, B])$

The system is inconsistent and has no solution.

Example 1.15

Find k, if the equations $x + 2y - 3z = -2$, $3x - y - 2z = 1$, $2x + 3y - 5z = k$ are consistent.

Solution:

The matrix equation corresponding to the given system is

$$\begin{pmatrix} 1 & 2 & -3 \\ 3 & -1 & -2 \\ 2 & 3 & -5 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \\ k \end{pmatrix}$$

$$A \qquad X = B$$

Augmented matrix $[A,B]$	Elementary Transformation
$\begin{pmatrix} 1 & 2 & -3 & -2 \\ 3 & -1 & -2 & 1 \\ 2 & 3 & -5 & k \end{pmatrix}$	
$\sim \begin{pmatrix} 1 & 2 & -3 & -2 \\ 0 & -7 & 7 & 7 \\ 0 & -1 & 1 & 4+k \end{pmatrix}$	$R_2 \to R_2 - 3R_1$ $R_3 \to R_3 - 2R_1$
$\sim \begin{pmatrix} 1 & 2 & -3 & -2 \\ 0 & -7 & 7 & 7 \\ 0 & 0 & 0 & 21+7k \end{pmatrix}$	$R_3 \to 7R_3 - R_2$
$\rho(A) = 2,\ \rho([A, B]) = 2$ or 3	

For the equations to be consistent, $\rho([A, B]) = \rho(A) = 2$

$\therefore \quad 21 + 7k = 0$

$7k = -21.$

$k = -3$

Example 1.16

Find k, if the equations $x + y + z = 7$, $x + 2y + 3z = 18$, $y + kz = 6$ are inconsistent.

Solution:

The matrix equation corresponding to the given system is

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 0 & 1 & k \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 7 \\ 18 \\ 6 \end{pmatrix}$$

$$A \qquad X = B$$

Augmented matrix $[A,B]$	Elementary Transformation
$\begin{pmatrix} 1 & 1 & 1 & 7 \\ 1 & 2 & 3 & 18 \\ 0 & 1 & k & 6 \end{pmatrix}$	
$\sim \begin{pmatrix} 1 & 1 & 1 & 7 \\ 0 & 1 & 2 & 11 \\ 0 & 1 & k & 6 \end{pmatrix}$	$R_2 \to R_2 - R_1,$
$\sim \begin{pmatrix} 1 & 1 & 1 & 7 \\ 0 & 1 & 2 & 11 \\ 0 & 0 & k-2 & -5 \end{pmatrix}$	$R_3 \to R_3 - R_2$
$\rho(A) = 2$ or 3, $\rho([A,B]) = 3$	

For the equations to be inconsistent

$$\rho([A, B]) \neq \rho(A)$$

It is possible if $k - 2 = 0$.

$$\therefore \boxed{k = 2}$$

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